

BARMER, IndiaWMO: **424350**Lat: **25.750N**Lon: **71.383E**Elev: **636**StdP: **14.36**Time Zone: **5.50 (IND)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.1	52.4	22.0	16.9	64.5	26.1	20.5	64.8	6.3	70.1	5.4	69.4	0.2	N/A	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.2	109.7	73.9	107.5	73.7	105.1	74.1	82.5	94.1	81.7	93.5	80.9	92.9	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	160.1	86.6	79.2	155.2	86.4	78.3	150.7	86.4	47.1	94.3	46.2	93.2	45.4	92.7	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	5.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			37.0	84.6	3.0	1.4	34.8	85.7	33.1	86.5	31.4	87.3	29.2	88.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	64.3	71.0	82.3	91.2	95.1	94.3	90.6	87.0	87.3	84.3	75.1	66.4	(6)	(7)
	DBStd	11.08	4.19	5.50	5.88	4.35	3.78	3.64	3.65	2.68	3.17	3.63	4.39	4.48	(7)	(8)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	(9)
	HDD65	109	66	8	0	0	0	0	0	0	0	0	0	35	(9)	(10)
	CDD50	11848	443	589	1001	1237	1398	1328	1259	1146	1119	1064	754	510	(10)	(11)
	CDD65	6481	44	177	536	787	933	878	794	681	669	599	304	79	(11)	(12)
	CDH74	95200	279	1587	7018	12778	15951	14594	12319	9573	9675	8061	2861	504	(12)	(13)
	CDH80	58936	26	516	3904	8626	11527	10294	7887	5173	5471	4375	1036	101	(13)	(14)
(14) Wind	WSAvg	1.2	0.6	0.9	0.9	1.5	2.5	2.3	2.2	1.2	1.0	0.4	0.3	0.3	(14)	(15)
(15) Precipitation	PrecAvg	11.8	0.0	0.2	0.1	0.2	0.4	1.1	3.3	3.5	2.1	0.3	0.1	0.0	(15)	(16)
	PrecMax	28.9	0.2	1.3	0.7	2.7	2.8	4.8	11.4	19.6	8.5	1.9	2.6	0.6	(16)	(17)
	PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	(18)
	PrecStd	6.1	0.1	0.3	0.2	0.5	0.7	1.4	3.0	4.3	2.5	0.5	0.5	0.2	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.2	93.8	103.9	110.4	113.4	113.2	106.9	99.0	101.5	99.8	93.4	86.8	(19)	(20)
		MCWB	62.8	65.2	69.9	71.3	73.5	75.1	79.3	79.6	77.4	72.9	67.0	64.6	(20)	(21)
	2%	DB	79.8	89.7	100.8	107.5	110.0	109.3	102.9	97.0	99.4	97.7	90.4	82.7	(21)	(22)
		MCWB	59.9	64.2	68.7	70.7	73.3	77.3	79.8	79.2	76.8	72.0	66.5	63.0	(22)	(23)
	5%	DB	77.3	86.1	98.1	105.2	107.9	106.2	100.6	95.3	97.5	95.8	87.9	79.3	(23)	(24)
		MCWB	58.6	63.0	67.1	70.4	72.9	77.6	79.3	78.9	76.9	71.3	65.9	61.1	(24)	(25)
	10%	DB	74.6	82.8	95.1	102.7	105.7	103.3	98.6	93.7	95.4	94.2	85.7	76.1	(25)	(26)
		MCWB	57.3	61.3	66.3	70.5	73.5	78.2	79.0	78.7	76.8	70.4	64.3	59.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	69.4	74.7	78.3	81.6	83.5	83.4	83.2	82.8	80.1	74.1	68.3	(27)	(28)
		MCDB	78.5	86.7	91.9	100.7	100.5	98.8	94.0	90.9	91.8	90.5	83.5	81.0	(28)	(29)
	2%	WB	62.7	67.6	71.8	76.2	80.0	82.1	82.3	81.8	81.4	78.7	71.9	64.9	(29)	(30)
		MCDB	75.7	84.0	93.8	98.4	99.0	97.7	93.6	90.2	90.7	88.6	83.2	77.9	(30)	(31)
	5%	WB	60.7	65.4	69.7	74.7	79.0	81.1	81.5	81.0	80.2	76.5	69.3	62.9	(31)	(32)
		MCDB	73.7	81.0	93.0	96.4	97.2	97.0	93.2	89.3	89.6	88.3	82.7	76.9	(32)	(33)
	10%	WB	58.7	63.4	68.1	73.3	78.0	80.4	80.8	80.4	79.3	74.8	67.2	60.8	(33)	(34)
		MCDB	72.2	78.7	91.1	95.2	96.3	96.5	92.8	88.7	88.8	88.5	82.3	74.4	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	21.8	22.3	21.5	22.2	18.5	14.9	12.1	14.7	18.3	18.9	20.1	(35)	(36)
		MCDBR	23.0	24.5	23.7	23.6	23.8	22.2	17.9	14.9	17.6	19.6	21.4	22.9	(36)	(37)
	5% WB	MCWBR	10.9	10.4	9.1	7.8	6.7	5.6	3.3	3.7	4.7	7.4	9.4	11.7	(37)	(38)
		MCWBR	20.9	20.2	21.7	21.8	22.7	18.8	14.6	12.2	13.7	16.4	17.3	21.6	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.407	0.434	0.489	0.560	0.620	0.693	0.735	0.656	0.566	0.507	0.478	0.430	(39)	(40)
		taud	2.070	2.019	1.885	1.761	1.681	1.600	1.561	1.708	1.864	1.885	1.905	2.008	(40)	(41)
	Ebn at Noon		262	265	257	242	227	210	201	217	233	239	236	247	(41)	(42)
		Edh at Noon	46	52	63	73	79	85	88	76	63	59	54	47	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		1358	1640	1977	2185	2262	2065	1705	1693	1816	1748	1410	1282	(43)	(44)
	RadStd		52	57	57	59	56	95	110	104	98	55	72	50	(44)	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page